

# OFFICIAL MINISTRY CONCEPTS & FORMULA COMPENDIUM

Authorized Ministry of Education Reference for General Secondary Exams

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 & EG-BACCALAUREATE EXAMINATIONS

## CURRICULUM OVERVIEW & AUTHORIZED SCOPE

The authorized official concepts compendium distributed to students inside ministerial examination halls, containing verified identities, integration forms, moment equilibrium rules, and physical constants.

### CORE CURRICULUM HIGHLIGHTS:

- Exact ministerial formula tables used in official examination halls
- Trigonometric and inverse trigonometric standard derivatives
- Integration forms, substitution catalogs, and reduction formulas
- Friction, couples, pulley equations, and momentum conservation laws

### APPROVED CHAPTER MODULES:

- |                                                                 |                          |
|-----------------------------------------------------------------|--------------------------|
| - Pure Mathematics Formula Reference (Algebra, Solid, Calculus) | pp. 1 44 (8 core units)  |
| - Applied Mathematics Formula Reference (Statics, Dynamics)     | pp. 45 88 (8 core units) |

## OFFICIAL PUBLICATION METADATA & VERIFICATION

Book Code: MOE-FORMULA-COMPENDIUM  
Pages: 88 Official Pages | Size: 9.4 MB  
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Verification URL: <https://moe.gov.eg/ar/elearning-content/>



# MINISTERIAL PREFACE & STATUTORY FRAMEWORK

## MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

### 1. Pedagogical Scope & National Objectives

The Official Ministry Concepts & Formula Compendium ( ) is the authorized reference document provided to students during the Thanaweya Amma and EG-Baccalaureate national examinations. It contains every statutory mathematical formula, identity, standard integral, differential equation model, and equilibrium law permitted during state examinations.

### 2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

|                                                                             |                                                                                       |                                                                                        |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>30%</b><br><b>Knowledge &amp; Recall</b><br>Definitions, laws & formulas | <b>40%</b><br><b>Understanding &amp; Application</b><br>Calculations, proofs & models | <b>30%</b><br><b>Higher Order Thinking (HOTS)</b><br>Analysis, synthesis & non-routine |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|

### 3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

### 4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

## SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: Authorized for All Senior Secondary Examination Sessions | Total Paper Marks: 60 Marks

| Chapter / Modular Title                                    | Weeks | Marks  | Weight | Curricular Focus                                             |
|------------------------------------------------------------|-------|--------|--------|--------------------------------------------------------------|
| Pure Mathematics: Algebra & Combinatorics Section          | 0 wks | 15 pts | 25.0%  | Counting, Binomial, Complex, Matrices                        |
| Pure Mathematics: Solid Analytical Geometry Section        | 0 wks | 15 pts | 25.0%  | 3D Lines, Planes, Vectors, Spheres                           |
| Pure Mathematics: Differential & Integral Calculus Section | 0 wks | 15 pts | 25.0%  | Trig/Log/Exp Derivatives, Standard Integrals, Areas, Volumes |
| Applied Mathematics: Statics & Dynamics Section            | 0 wks | 15 pts | 25.0%  | Friction, Moments, Equilibrium, Newton Laws, Work & Energy   |

## INTERDISCIPLINARY COGNITIVE BRIDGES

|                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Physics (Mechanics &amp; Electromagnetism)</b></p> <p>Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.</p> |
| <p><b>Engineering Sciences (Civil, Mechanical, Electrical)</b></p> <p>Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.</p>               |
| <p><b>Computer Science &amp; Algorithmics</b></p> <p>Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.</p>      |

## CHAPTER SPECIFICATION: SECTION 1: PURE MATHEMATICS FORMULA REGISTRY

### A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Immediate recall and error-free substitution into all ministerial standard formulas.
- [+] Verify algebraic constraints before applying specialized formulas.

### B. Statutory Theorems & Analytical Principles

All formulas in this compendium are verified by the National Center for Examinations and Educational Evaluation (NCEEE).

### C. Standard Mathematical Expressions & Operational Formulas

- >  $nPr = n! / (n - r)!$
- >  $nCr = n! / [r! (n - r)!]$
- >  $nCr / nC(r-1) = (n - r + 1) / r$
- >  $T_{(r+1)} = nCr a^{n-r} b^r$
- > Euler Identity:  $e^{(i \theta)} = \cos \theta + i \sin \theta$
- > De Moivre:  $(\cos \theta + i \sin \theta)^n = \cos(n \theta) + i \sin(n \theta)$
- > Omega:  $\omega^3 = 1, 1 + \omega + \omega^2 = 0, \omega - \omega^2 = \pm i \sqrt{3}$
- >  $A^{-1} = (1 / |A|) \text{Adj}(A)$

### D. National Examination Pitfalls & Common Student Errors

- [!] Writing formulas from memory rather than verifying exact indices and signs against this official sheet.

## CHAPTER SPECIFICATION: SECTION 2: APPLIED MATHEMATICS FORMULA REGISTRY

### A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

[+] Direct lookup of mechanics equilibrium and dynamics principles during timed national examinations.

### B. Statutory Theorems & Analytical Principles

Statics and dynamics equations verified under SI and gravitational metric standards.

### C. Standard Mathematical Expressions & Operational Formulas

- >  $F_s = \mu_s \cdot R$  (limiting friction)
- >  $R' = R \sec \lambda$ ,  $\tan \lambda = \mu_s$
- >  $M_O = r \times F$ ,  $|M_O| = F \cdot L$
- > Parallel Forces:  $F_1 \cdot x_1 = F_2 \cdot x_2$
- > Center of Mass:  $X_G = \frac{\sum(m_i \cdot x_i)}{\sum(m_i)}$
- >  $v = \frac{dx}{dt}$ ,  $a = \frac{dv}{dt} = v \left(\frac{dv}{dx}\right)$
- >  $F = m \cdot a$ ,  $W = \int F \, dx$
- >  $T = \frac{1}{2} m v^2$ ,  $\Delta T = W$
- >  $P = F \cdot v$ ,  $1 \text{ HP} = 735 \text{ W} = 75 \text{ kg}\cdot\text{wt}\cdot\text{m/s}$

### D. National Examination Pitfalls & Common Student Errors

[!] Using non-matching units when combining statics moments and dynamics forces.

# OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

| PURE MATH FAST REFERENCE                                                  |
|---------------------------------------------------------------------------|
| $\cos^2 x + \sin^2 x = 1$                                                 |
| $1 + \tan^2 x = \sec^2 x$                                                 |
| $1 + \cot^2 x = \csc^2 x$                                                 |
| $\sin(2x) = 2 \sin x \cos x$                                              |
| $\cos(2x) = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$        |
| $\frac{d}{dx} [\tan x] = \sec^2 x, \frac{d}{dx} [\sec x] = \sec x \tan x$ |
| $\text{Integral } (f'(x) / f(x)) \, dx = \ln f(x)  + C$                   |
| $\text{Integral } e^{(k x)} \, dx = (1/k) e^{(k x)} + C$                  |

| MECHANICS FAST REFERENCE                                   |
|------------------------------------------------------------|
| $F_{\text{max}} = \mu_s R$                                 |
| $M_0 = r \times F$                                         |
| Equilibrium: $\sum F_x = 0, \sum F_y = 0, \sum M = 0$      |
| Elevator: $N = m(g \pm a)$                                 |
| Atwood: $a = (m_1 - m_2)g / (m_1 + m_2)$                   |
| Impulse $I = F \Delta t = m \Delta v$                      |
| Work-Energy: $\frac{1}{2} m v^2 - \frac{1}{2} m v_0^2 = W$ |

# NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

## 1. Structure of the National Examination Paper

|                          |                                                                                    |
|--------------------------|------------------------------------------------------------------------------------|
| Paper Duration:          | Authorized for All Senior Secondary Examination Sessions                           |
| Total Scaled Marks:      | 60 Marks                                                                           |
| Section I (Objective):   | Multiple Choice Questions (MCQ) with 4 options; single best answer.                |
| Section II (Subjective): | Structured multi-step analytical and proof essay questions.                        |
| Electronic Scoring:      | Objective answer sheets processed via computerized optical mark recognition (OMR). |
| Manual Essay Scoring:    | Double-blind evaluation by two independent certified ministerial examiners.        |

## 2. Standard Candidate Exam Directives

- Keep this compendium accessible on your exam desk at all times.
- No handwritten annotations or additional sheets may be attached to this official booklet.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION  
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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